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ET. 8 SCM

Printed Pages – 8

328873 (28)

BE (8th Semester)

Examination, April - May, 2013

Branch : ET & T

INFORMATION THEORY & CODING (PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) All questions carry equal marks.

UNIT - I

Attempt any two parts from (b), (c) and (d) :

Q. 1. (a) Calculate extended entropy $H(s^2)$ of the source symbol whose probabilities are :

0.2 0.3 and 0.50

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(b) Draw and explain the model of digital communication system.

(c) A 4 KHz channel has a signal to noise ratio 24 dB. Calculate :

(i) Maximum capacity of the channel

(ii) Assuming constant transmitting power calculate the maximum capacity, when channel bandwidth is halved.

(iii) Assuming constant transmitting power calculate the maximum capacity, when channel bandwidth reduced to a quarter of the original value.

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(d) Explain binary symmetrical channel and derive the expression for the mutual information.

UNIT - II

Attempt any two parts from (b), (c) and (d) :

Q. 2. (a) Define block code and convolution code.

(b) A (6, 3) code is generated according to the generator matrix. The receiver receives Y = 100110. Determine the corresponding data word if the channel is a binary symmetric channel and maximum likelihood decision is used.

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$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

(c) The impulse response of the input to adder output path and input bottom adder output path of convolution encoder is {1, 1, 1} and {1, 0, 1} respectively and message sequence is {1 0 1 1 0}. Calculate the convolution encoder sequence.

(d) Explain in details the single error correcting hamming codes with suitable example.

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UNIT - III

Attempt any two parts from (b), (c) and (d) :

Q. 3. (a) What is run length coding ? Explain with suitable example.

(b) Explain the advantages of variable length coding over fixed length coding, also explain the applications of variable length coding ?

(c) Explain how the data bytes are compressed also explain for the data compression which compression method is better lossless or lossy and how, justify your statement ?

(d) Draw and explain CCITT group-3 type image compression with suitable example.

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UNIT - IV

Attempt any two parts from (b), (c) and (d) :

Q. 4. (a) Explain H.264 compression technique ?

(b) Explain how the MPEG compression technique is employed, also explain the different stages involved for the MPEG compression.

(c) Explain any two DCT methods which are supported to MPEG compression.

(d) Draw and explain in details the motion estimation techniques in MPEG compression.

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UNIT - V

Attempt any **two** parts from (b), (c) and (d) :

Q. 5. (a) Define passive wiretapping and cryptography.

(b) Find all solution using cryptography approach to the equation 6 X mod 10=8 in the range [0, 9].

(c) Encrypt a messages given below using permutation method for a period of 5 characters in the permute sequence of 3 5 1 4 2. Ignore spaces

(i) INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

(ii) BHILAI INSTITUTE OF TECHNOLOGY RAIPUR

(iii) THE GREAT SACHIN TENDULKAR

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(d) The encrypt message is "T T N A A P T M T S U O A O D W C O I X K N L Y P E T Z" is generated by reading column in the order of 3 4 2 1 5 6 7. Find out the original plain text.

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